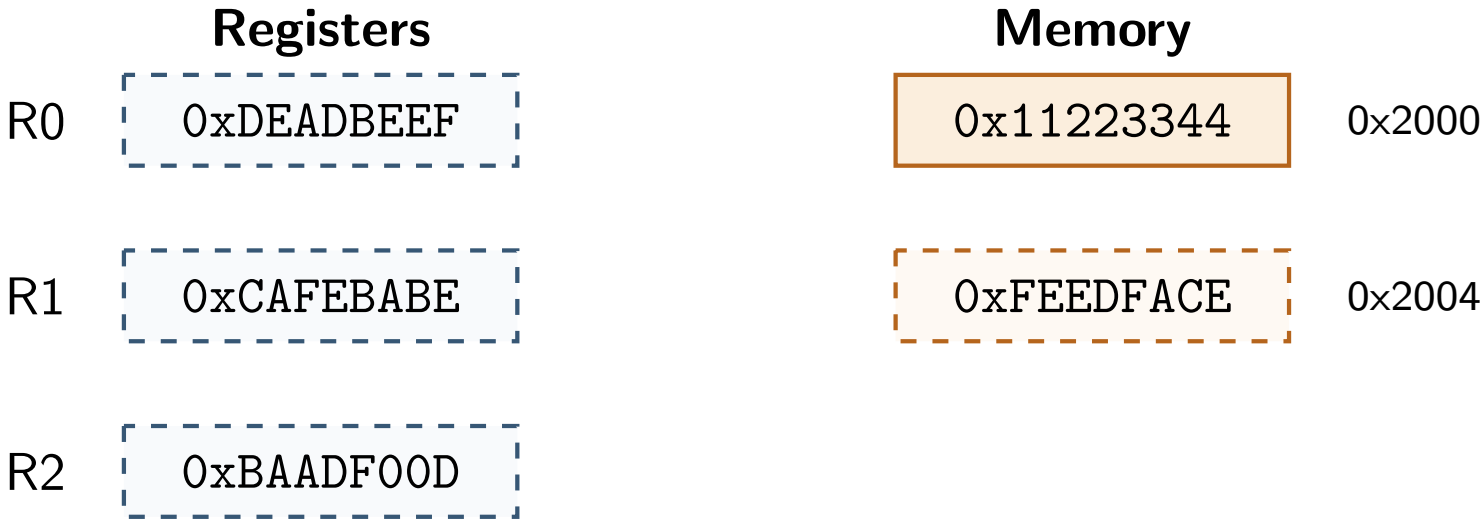
Data Copy: LDR / LDR / LDR / STR

```
ldr r0, =0x2000
ldr r1, =0x2004
ldr r2, [r0]
str r2, [r1]
```

Initial state: R0, R1, R2 hold garbage; [0x2000] already holds the data to copy.

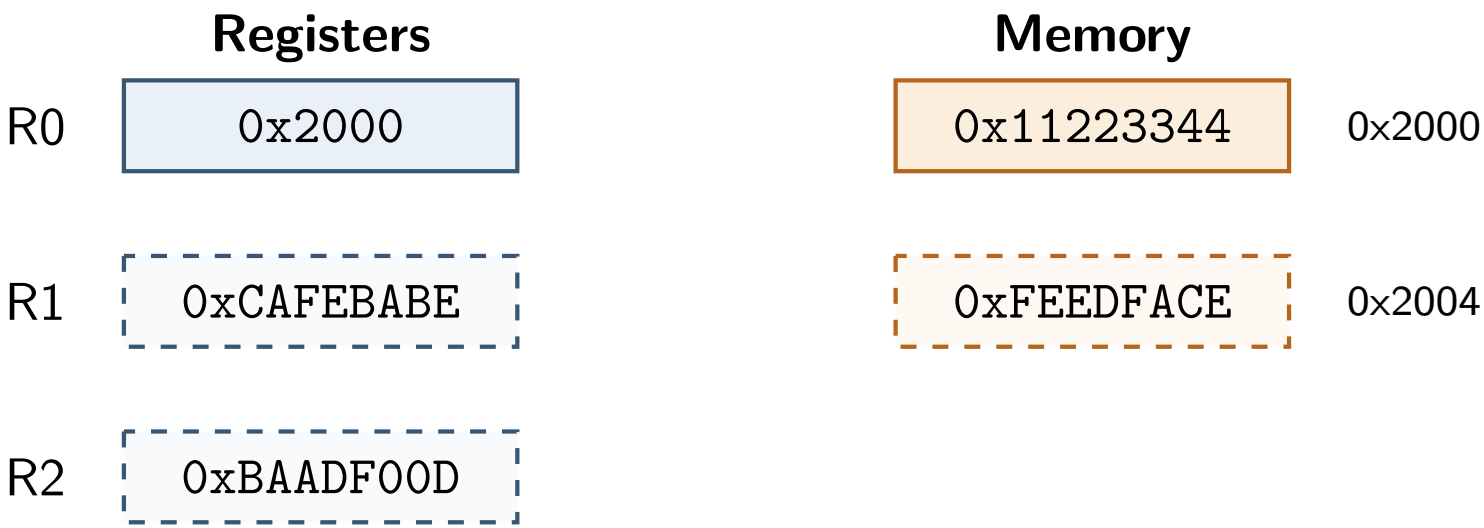


dashed arrow = address used for addressing solid arrow/solid arrow = data value moved

Data Copy: LDR / LDR / LDR / STR

```
ldr r0, =0x2000
ldr r1, =0x2004
ldr r2, [r0]
str r2, [r1]
```

Step 1: R0 is loaded with the source address 0x2000.

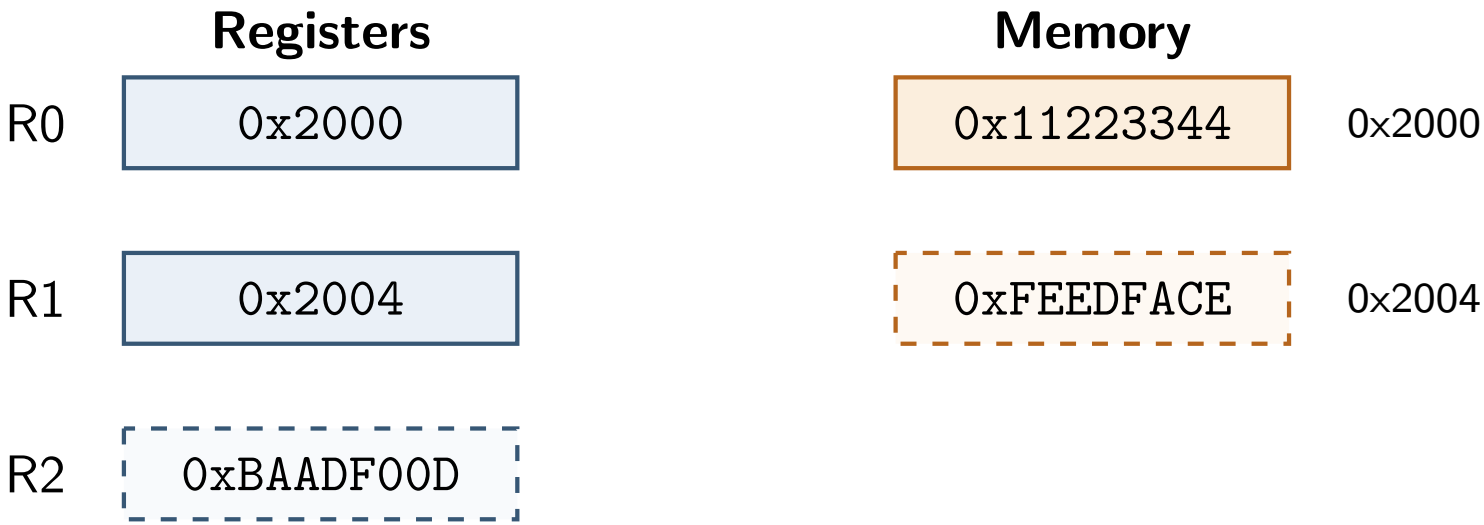


dashed arrow = address used for addressing solid arrow/solid arrow = data value moved

Data Copy: LDR / LDR / LDR / STR

```
ldr r0, =0x2000
ldr r1, =0x2004
ldr r2, [r0]
str r2, [r1]
```

Step 2: R1 is loaded with the destination address 0x2004.

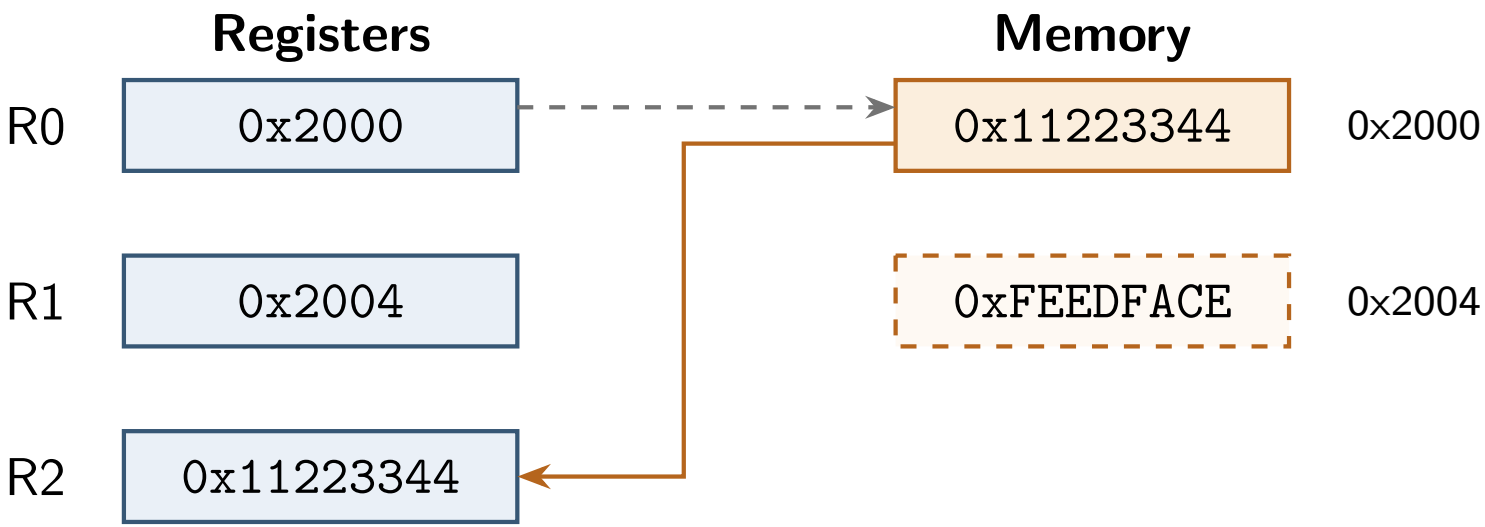


dashed arrow = address used for addressing solid arrow/solid arrow = data value moved

Data Copy: LDR / LDR / LDR / STR

```
ldr r0, =0x2000
ldr r1, =0x2004
ldr r2, [r0]
str r2, [r1]
```

Step 3: the word at address [R0] is read from memory into R2.

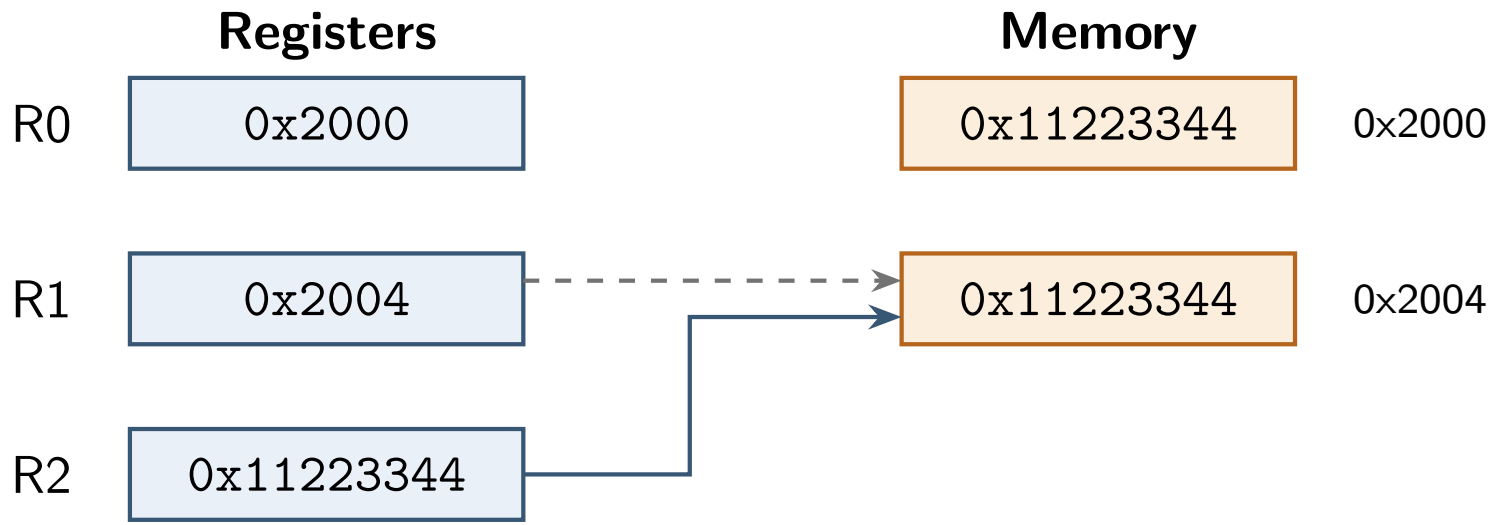


dashed arrow = address used for addressing solid arrow/solid arrow = data value moved

Data Copy: LDR / LDR / LDR / STR

```
ldr r0, =0x2000
ldr r1, =0x2004
ldr r2, [r0]
str r2, [r1]
```

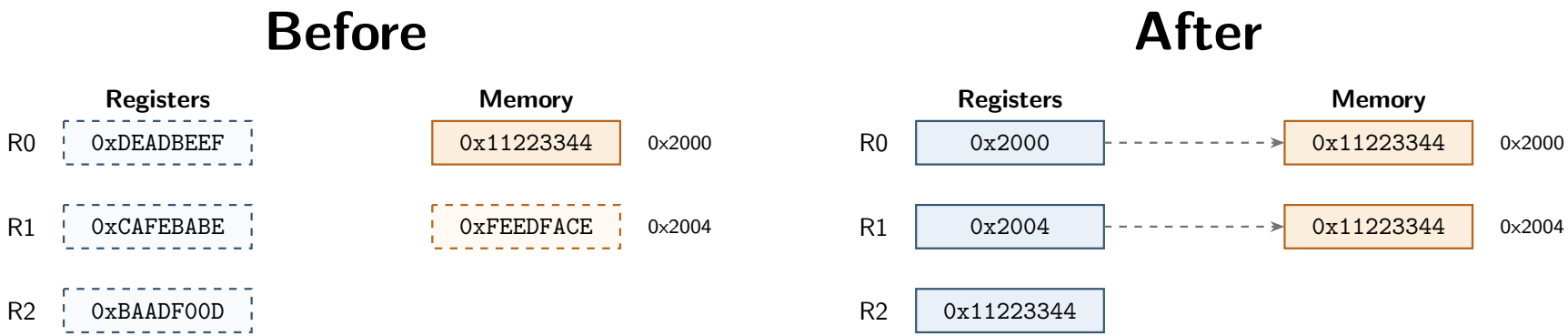
Step 4: R2 is written back out to memory at address [R1] — the copy is complete.



dashed arrow = address used for addressing solid arrow/solid arrow = data value moved

Data Copy: Before / After

```
ldr r0, =0x2000
ldr r1, =0x2004
ldr r2, [r0]
str r2, [r1]
```



dashed arrow = address used for addressing